

Introduction to GSE risk transfer deals

Freddie Mac recently announced a new deal to help shed the credit risk of the mortgages it guarantees to the private sector, by issuing unsecured debt whose cash flows mimic a first loss piece on the underlying reference collateral. We believe that if the GSEs sell the risk on all their ongoing issuance, we could see a program that has annual issuance of about \$20-25bn in credit tranches, though it may be a while before this happens.

- The structure effectively creates credit-linked notes issued by Freddie Mac as unsecured debt, with the balance of the notes affected by prepays and defaults on the underlying reference pool. The severities are fixed in three slabs of cumulative defaults.
- The structure also accounts for modifications, so that the issued notes do not take a loss for balance forgiveness modifications. Adjustments are also made for rep and warranty repurchases by reversing losses. Finally, the notes are called at the end of ten years, so Freddie retains the risk of defaults beyond that point.
- Defaults are defined as the earlier to occur of 180+ day delinquency (MBA-style) or termination of the loan through short-sale, REO sale, foreclosure note sale or deed in lieu. As a result, modelling performance is simplified to mainly projecting the current to delinquent and current to prepay rolls. Other rolls that will matter are modifications in the first six months and rolls to short-sale or REO sale in the first six months of delinquencies.
- This structure protects the taxpayer from certain risks by transferring them to the private market, provides the GSEs a way to achieve true risk-based pricing of the guarantee fee and is likely to be appealing to a range of political camps. Importantly, it achieves all of these without affecting the agency MBS market.
- The reference loan pool for the first transaction comprises loans acquired by Freddie Mac in the Q3 12. The loans have original LTVs between 60% and 80% and have fully documented incomes. The tail characteristics of the pool are very well controlled compared to historical Freddie Mac vintages. Furthermore, the pool has experienced about 9-10% increases in home prices since origination.
- We expect the cumulative defaults on the pool over the first ten years to be about 82bp in our base case and to increase to about 1.7% in our severe stress scenario. If home prices drop 20% over the next three years before recovering, we project defaults on the pool over 10 years to be about 3.5%.

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- In terms of valuation, the M1 looks attractive on an unleveraged basis compared to similarly leveraged tranches in the CMBS new issue and legacy space. With leverage, the returns look even more enticing. On the M2, the yields can drop off for much smaller shocks in defaults. With leverage, however, the M2 also looks attractive, but we would remain wary of MTM risk and any turn in housing expectations.
- We believe that if this issuance from the GSEs evolves into a program, then it should help in price-discovery of the credit risk on well-defined, liquid and large cohorts of loans. This would provide a series of benchmarks against which private-label deals could price new issuance.
- It could also provide Congress with an alternative approach to housing finance. Congress could still transition away from the GSEs, but could do so at a slower pace that does not negatively affect the housing markets. After all, if the taxpayer is being protected from the credit risk of new loans, then the urgency to transition away should be reduced.

Background

In March 2013, the FHFA released its updated Conservatorship Strategic Plan for 2013 with specific goals on contracting GSEs presence in the markets. As part of this the plan, the FHFA called for each of the enterprises to demonstrate the viability of risk transfer transactions involving single-family mortgages with at least \$30bn of unpaid principal balances in 2013 (see *GSEs shedding credit risk*, 18 November 2011, for our article outlining a similar program). The first of these transactions was announced by Freddie Mac last week. In this article, we look at the details of the deal and the possible path of the program more broadly. In Section I, we look at the detailed structure and cash flow waterfall of the deal. In Section II, we examine the deal collateral and dig into the tail risks of the pool. We also compare the pool with historical Freddie and jumbo cohorts to find comparables to estimate likely defaults on the pool. In Section III, we look at valuation of the notes on an unleveraged and leveraged basis and finally in Section IV, we look at the possible size of this market, likely sources of demand and the likely benefits of such a program.

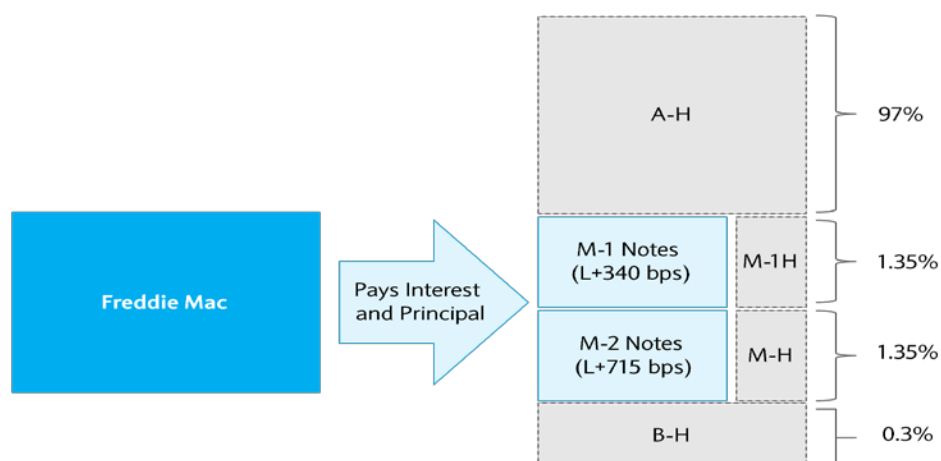
Section I: Deal structure and cash flow waterfall

The Freddie Mac deal is called STACR 2013-DN1 and could be the first among many deals over the coming months and years. We believe that with the pressure staying on GSEs to reduce their footprint in the mortgage market such deals will become the norm from both GSEs over the coming years. The bonds that are issued are structured similar to floating rate credit-linked notes. They are structured so that they can be considered Freddie Mac unsecured debt instruments for tax purposes. Figure 1 shows the transaction structure in broad terms.

Reference tranches

The reference pool in this case is a pool of about \$22.6bn of Freddie wrapped loans that were acquired by Freddie Mac between July 1, 2012 and September 30, 2012. The structure allocates scheduled and unscheduled principal as well as losses, from predefined credit events at predefined severities, to the reference tranches shown in the structure diagram (Figure 1). The notes that are issued use the corresponding reference tranches as a guide for cash flows and write-downs. The senior-most reference tranche A-H has 3% credit support and Freddie Mac retains the credit risk on this tranche. The junior-most tranche is 0.3% thick, and Freddie Mac retains the risk on that tranche completely as well. As issued, the M1 and M2 notes form about 82% of the combined M-1/M-1H and M-2/M-2H reference tranches, respectively, with Freddie intending to hold onto at least 5% of risk on these reference tranches. The M2 tranche has an attachment point at 0.3% and detaches at 1.65%, while the M1 has 1.65% support and detaches at 3%. On this deal the M1 priced at L+340bps and the M2 priced at L+715bps.

FIGURE 1
Transaction Structure



Note: M-1 and M-2 notes issued were sized at \$250mn each. The unissued M-1H and M-2H tranches were sized at \$54.88mn each. Effectively Freddie Mac sold 82% of the entire M-1/M-2 tranches. Source: Barclays Research

Credit events and applicable severities

The deal defines certain credit events upon occurrence of which the balance of the mezzanine and subordinate reference tranches is reduced. The loss allocated to the tranches is equal to the product of the unpaid balance (UPB) affected by the credit event times the applicable severity. The deal defines the first-to-occur of the following as a credit event for this purpose for each loan in the reference pool:

- loan is 180 days or more delinquent (MBA), regardless of any deferral granted in connection with natural disasters or if the loan is performing in a trial modification
- loan on which a short-sale is settled
- the mortgage note is sold to a third party during foreclosure
- deed in lieu of foreclosure is executed
- an REO acquisition occurs

Reversal of credit events

A credit event can be reversed if it is subsequently determined, through Freddie Mac's quality control process, that the loan in question had an underwriting defect or if the data were misreported. An underwriting defect is recognized and used to reverse credit events in the following circumstances for loans in which Freddie has previously identified material violation of underwriting guidelines:

- the reference obligation is repurchased by the related seller or servicer
- alternate remedy agreed upon by Freddie Mac
- Freddie Mac elects to waive enforcement of a remedy

In general, the credit events can be thought of as defaults for the purpose of this transaction.

Applicable severities

Finally to convert the credit event to a loss that is passed to the notes, the structure defines certain applicable severities. These severities vary based on the cumulative net credit events

until that payment date. “Cumulative net credit event percentage” is measured as a percentage of the reference obligations at issuance and effectively measures the cumulative net default as a percentage of the original balance.

Applicable severities follow the schedule shown in Figure 2. So until the cumulative defaults reach 1%, a 15% severity is applied, after which a 25% severity is applied until 2% cumulative defaults and then a 40% severity is applied beyond that. For reversed credit events, the same schedule is applied based on the cum default rate at the point of reversal.

FIGURE 2
Applicable severities

Cumulative Net Credit Event Percentage	Applicable Severity
0 to 1%	15%
1 to 2%	25%
More than 2%	40%

Source: Offering Circular, Barclays Research

Treatment of modifications

Loans will remain in the reference pool if they undergo a temporary or permanent modification as long as they do not exit the pool through either a credit event or a full prepay (or because of an underwriting or eligibility defect). Naturally if the loan is more than 180 days delinquent at modification then, since a credit event would have occurred already, such a loan would exit the pool through the usual credit event channels. As a result of the modification, if the balance of the reference obligation is increased, then this will be treated as a negative prepayment. If the balance is reduced, then that will be treated as Unscheduled Principal (i.e., if FHFA/ Freddie decides to ever pursue forgiveness modifications then the amount forgiven will not cause losses to the notes but will be treated as a partial prepayment).

Deal has cumulative default and credit enhancement triggers

So far we have described how credit events and losses are computed but have not described how these are applied to the tranches. The final piece of the puzzle before we describe the cash flows is to understand triggers that apply. The deal uses two triggers to determine how principal allocations work. The first trigger is based on cumulative net credit event percentage and the schedule is shown in Figure 3. The trigger is passed as long as the cumulative net default rate does not exceed the thresholds shown for the payment date range corresponding to the threshold. The second trigger test is based on the credit enhancement for the A-H reference tranche and passes as long as the enhancement is equal to or greater than 3%.

FIGURE 3
Cumulative net credit event trigger

Payment date ranges	Percentage cumulative net credit event threshold
Aug 2013 to Jul 2014	0.25%
Aug 2014 to Jul 2015	0.50%
Aug 2015 to Jul 2016	0.75%
Aug 2016 to Jul 2017	1.00%
Aug 2017 to Jul 2018	1.25%
Aug 2018 to Jul 2019	1.50%
Aug 2019 to Jul 2020	1.75%
Aug 2020 to Jul 2021	2.00%
Aug 2021 to Jul 2022	2.25%
Aug 2022 and beyond	2.50%

Source: Offering Circular, Barclays Research

Cash flow and loss allocation

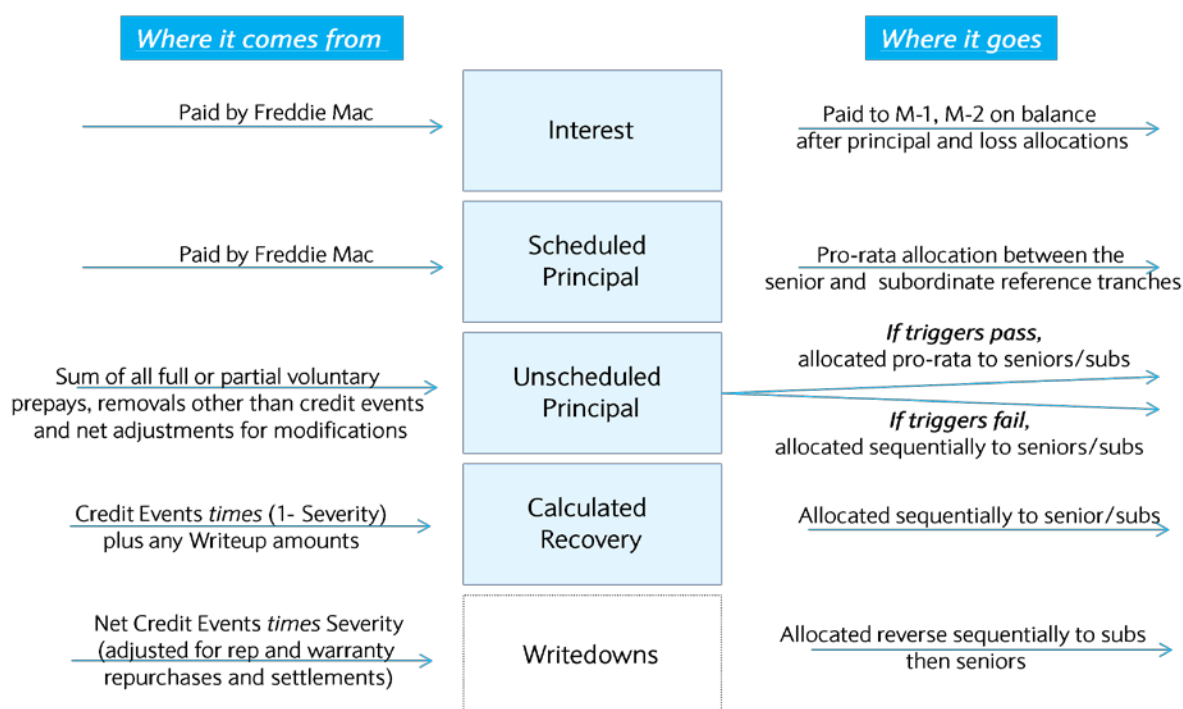
Since cash flows of the notes are paid directly from Freddie Mac, there is no dependence of the interest cash flows on the actual underlying interest cash flows. The notes simply get paid the accrued interest based on the outstanding balance of the notes directly from Freddie Mac.

Principal cash flows are more closely dependent on the reference pool and are divided into three parts, each of which are then allocated across the reference tranches based on whether the triggers pass or fail. The parts of principal collections are defined below:

- **Scheduled principal** – the total scheduled principal actually collected on the loans by the servicer (i.e., not including any advances on delinquent loans that are not yet credit events)
- **Unscheduled principal** – sum of all partial prepayments, full prepayments or removals for other reasons (except for credit events) and any net adjustments in principal due to modifications
- **Principal recovery** – the UPB of the credit event loans less the implied loss based on the applicable severity shown in Figure 2

Figure 4 shows a schematic of the sources and allocation of various components of the deals cash flows. If both the triggers are passing then the Senior (A-H) and Subordinate (the M-1, M-1H, M-2, M-2H & B-H reference tranches together) get pro-rata allocations of the scheduled and unscheduled principal amounts. If triggers are failing, then only the scheduled principal is allocated pro-rata but the unscheduled principal is all allocated to first reduce the balance of the A-H tranches. Regardless of the trigger status, the Principal Recovery is always fully allocated first to reduce the A-H balance. Once the A-H tranche is paid down then any excess that would have been allocated to the senior is allocated to the subordinate tranches instead.

FIGURE 4
Cash flows and losses, sources and allocation



Source: Barclays Research

Triggers will toggle, locking out the subs

The enhancement trigger can toggle between passing and failing because when trigger fails, additional principal gets allocated to the A-H tranche, building up enhancement and possibly leading to the triggers passing again. Technically, since the cumulative default trigger is a step function, each August it can toggle as well. So to begin with, the subordinates will receive pro-rata share of prepaids but they can all get locked out if the triggers fail.

Sequential principal allocation among the notes

The principal allocated to the subordinates is allocated sequentially, first to the M-1/M-1H, then to M-2/M-2H and finally to B-H reference tranches. Within the M-1/M-1H tranches the allocation is pro-rata and similarly for the M-2/M-2H tranches. This means that the M-2/M-2H and B-H tranches are locked out until the M-1/M-1H tranche balances are reduced to zero. Freddie Mac pays out as principal cash flow to the notes, any allocation that goes to the M-1 or M-2 reference tranches on any payment date.

Loss allocation – Write-downs allocated reverse sequentially, write-ups sequentially

Allocation of write-downs (net credit event balance times applicable severity, if positive) is in simple reverse sequential order with B-H allocated all amounts until its balance is zero followed by pro-rata allocations to M-2/M-2H, followed by allocations to the M-1/M-1H tranches and finally to the A-H reference tranche if all the mezzanine certificates get written down to 0 balance. If, on the other hand, in any month the net credit event amount is negative (meaning more reversals than new credit events), then the reversals would translate into a write-up for the bonds that have been previously allocated write-downs. The allocation of the write-ups is in the opposite order of write-down, so first to A-H if any, then to M-1/M-1H and so on. The bonds can only be written up to the extent that they have had write-downs.

Mandatory call at the end of 10 years

The notes have a hard maturity date at the end of ten years from issuance. If the balance of these notes remains outstanding at the end of 10 years, Freddie Mac will pay par on any remaining notional at that point. This means that any collateral losses that happen more than 10 years after issuance do not affect the notes and any future losses are borne by Freddie Mac.

Other miscellaneous provisions and factors

The deal has a few other features that may affect overall cash flows and valuations under certain specific circumstances. We look at some of the more important ones below.

Early redemption option

Besides the 10 year mandatory call that we mention above, if the reference pool balance falls below 10% of the cut-off balance, then Freddie Mac has the option of calling the outstanding notes at par plus any accrued and unpaid interest.

Rep and warranty settlements with originators

If Freddie Mac enters into a broad settlement on the loans in the reference pool, then the benefit of that settlement will accrue to the notes to the extent the benefit pertains to the loans in question. The benefit will be used to reverse the net credit events occurring until the settlement date and beyond on the reference obligations that are part of the settlement. This is done by reversing such credit events and writing up the notes by the product of applicable severity and the credit event balance. However, the cumulative benefit (total write-ups as a result of the settlement) cannot exceed the settlement amount allocable to the loans in the pool out of all the loans that form part of the settlement. For instance, if \$100mn of loans by an originator are part of the settlement and only \$10mn of that is in the

reference pool, then assuming the settlement is across all loans (and not at a loan level), the maximum benefit that can apply to write-up the tranches will be equal to 10% of the total settlement benefit accruing to Freddie Mac.

Rights upon EOD – no access to collateral

One additional factor for investors to consider is that while the structure mimics the cash flows of a mezzanine securitized tranche, the notes are not directly linked to the underlying reference pool and in no circumstances give direct access to the loans in the reference pool. This is different than in the case of covered notes, where the issuer default would allow investors to have an additional lien on the loans. As a result, in the unlikely scenario that Freddie Mac fails to pay principal or interest on the notes or goes through a bankruptcy, insolvency or similar proceeding (conservatorship of Freddie Mac would not be considered an Event of Default), there is no recourse to the underlying loans that is directly available to the note holders. They will receive recovery on par with other unsecured note holders (agency debentures) in that unlikely scenario.

Section II: Collateral and performance expectations

The reference pool behind this deal comprises about \$22.6bn of loans acquired by Freddie Mac in Q3 2012 and securitized by January 31, 2013. The loans met the following eligibility criteria:

- Fully amortizing 1-4 unit first lien, fixed 30y mortgages
- Originated after April 1, 2012
- Outstanding as of July 2, 2013
- Has no known underwriting defects as of May 31, 2013
- Has never been reported 30 days or more delinquent since purchase as of May 31, 2013.
- Originated with full documentation (either Streamlined Accept or Standard)
- Loans do not have mortgage or pool level insurance policies.
- Original LTVs are greater than 60 and less than or equal to 80
- Not subject to recourse or other credit enhancements
- Not originated under HARP/Home Possible or other affordable mortgage programs or under FHA/VA/RHS programs
- UPB at cut-off is greater than \$5,000

Tail characteristics are very well contained

When evaluating the credit of such a reference pool for valuing the bottom 3% credit tranches, as in this case, average characteristics are less useful. What investors should care about more are the worst tails of the distributions across various characteristics that affect performance. Figure 5 shows the distribution of the reference pool across a set of characteristics with the tails highlighted in red. Overall the pool is very clean from both an average and tail perspective. Less than 3% of the pool is below 680 FICO and a similar number is above or equal to 90% CLTV. On the DTI front as well the proportion above 50 DTI is less than 1%.

FIGURE 5

Characteristics distribution of STACR 2013-DN1

FICO Buckets	% Balance	OCLTV bucket	% Balance	DTI Bucket	% Balance
<680	3%	61-70	22%	<=30	43%
681-720	10%	71-79	35%	31-40	33%
721-780	44%	=80	38%	41-49	23%
>780	44%	81-89	2%	50 or worse	1%
		>=90	3%		

Source: Barclays Research

Loans with layered tail risks absent from the pool

The loans most at risk of default are those with layered tail risks (loans with more than one of the above three tail risks). Figure 6 shows that in terms of layered risk, the pool is extremely clean. The numbers of loans with high CLTVs and low FICO, for instance, is about 10bp. Similarly, layered risk measures across other dimensions also reveal fairly low concentrations of these loans.

FIGURE 6

Layered risks are very limited

FICO	CLTV			DTI			CLTV	DTI		
	<=80	81-89	>=90	<=40	41-49	>=50		<=40	41-49	>=50
>720	82%	2%	3%	67%	19%	1%	<=80	71%	22%	1%
681-720	10%	0.2%	0.4%	7%	3.2%	0.1%	81-89	2%	0.4%	0.0%
<680	2.4%	0.0%	0.1%	1.7%	0.8%	0.0%	>=90	2.8%	0.7%	0.0%

Source: Freddie Mac, Barclays Research

Historical characteristics and performance of Freddie Mac loans

Freddie Mac recently released data on its guarantee fee (g-fee) portfolios going back to the 1999 vintage. We show below the original characteristics of these cohorts along with cumulative defaults performance and home price experience on the original pool in Figure 7. Since the STACR deal is unaffected by credit events that occur more than 10 years after origination, we show performance and home prices for the shorter of 10 years since origination or until date since issuance.

Performance of late 90s-early 2000s likely a cap for expected cumulative defaults

This deal is being issued about a year after origination, so technically it may be better to look at 1-11 year defaults, but for simplicity we stick with 0-10 year cumulative defaults. The numbers are not meaningfully different for these vintages in any case, since the cumulative default in the first year is small due to the six-month dq definition and cumulative defaults in the 11th year are small due to the small factor outstanding at that point for most cohorts. Future deals may have a shorter gap between agency MBS issuance and credit deal issuance so 0-10 may be more meaningful in the longer run. To get closer to STACR collateral, we have limited our loan pool to 60-80% LTV at origination and eliminated loans with mortgage insurance. We do not eliminate or make any adjustments for any repurchases or settlements on the loans that default. Thus, our numbers shown below can be thought of as gross cumulative default numbers.

We define cumulative defaults for this purpose using the same definition as for credit events under the STACR transaction and find that, for the late 1990s and early 2000s vintages, the cumulative defaults are of the order of 1-1.2% for loans that have experienced a positive annual rate of home price growth of about 1.5-4% (1999-2001 vintages). Obviously, once

we start getting to the 2004-2008 vintages, cumulative defaults have been much worse. Looking across to the average and tail characteristics for each vintage, as shown in Figure 7, we find that the defaults correlate reasonably well with the worse housing experience and also with worse average and tail characteristics.

FIGURE 7

Historical default performance, home price experience, average and tail characteristics of Freddie cohorts

Vintage	Shorter of 10y since Orig. or until date			Orig. FICO Score		Orig. CLTV		Back-end DTI	
	Cum Def	Cum. HPA	Ann. HPA	Average	% FICO<680	Average	% OCLTV=>90	Average	% DTI=>50
1999	1.2%	46%	3.8%	716	24%	76	1%	32	11%
2000	1.0%	29%	2.6%	719	23%	77	8%	35	9%
2001	1.1%	15%	1.4%	720	22%	77	7%	33	8%
2002	1.5%	10%	1.0%	721	21%	77	9%	34	10%
2003	2.2%	8%	0.8%	726	19%	76	9%	33	10%
2004	4.0%	-2%	-0.2%	720	23%	78	13%	35	13%
2005	8.2%	-14%	-1.9%	724	22%	78	14%	37	16%
2006	11.2%	-17%	-2.6%	725	22%	80	20%	38	18%
2007	12.1%	-15%	-2.6%	726	22%	80	21%	38	19%
2008	6.7%	-4%	-0.9%	741	12%	77	7%	38	21%
2009	0.8%	6%	1.4%	762	3%	76	3%	34	9%
2010	0.2%	8%	2.5%	762	3%	76	3%	34	4%
2011	0.1%	13%	6.2%	763	3%	76	3%	34	2%
STARC-2013-DN1				766	2.6%	76	3.4%	32	1.3%

Note: We define defaults as 180+ or terminations similar to the STARC definitions. Data shown for a 1.6mn loan sample of Freddie wrapped loans excluding those with mortgage insurance and only including loans with original LTVs between 60% and 80%. Each vintage sample has about 100-130K loans in the sample. Cumulative defaults, Cumulative and Annualized HPA shown for 10 years since originations or until date, whichever is shorter. Performance data are cut off as of June 2012.

Source Freddie Mac, Corelogic, Barclays Research

STACR much cleaner than 99/early 2000s; but these cohorts had the benefit of higher prepaes

The last line in Figure 7 also shows the STACR collateral characteristics as a comparison. Compared to the late 1990s and early 2000s, the STACR deal has on average 40-45 points higher FICO, similar average CLTV at origination and slightly lower DTI. However, the older vintage pools were more bar-belled, with much worse tails across characteristics. For example, even looking at LTV and DTI tails where the averages are comparable, we see that while STACR has about 2-3% of the balance in those tails, the tail balance in the earlier vintages was close to high-single, low-double digits. For FICO also, we have a much better tail in the reference pool than in the 1999-2002 vintages. These vintages did experience fairly high prepaes initially and a big increase in home prices which could have cushioned the remaining loans from the big declines in home prices experienced between 2006 and 2010. So it is possible that if these vintages had been 50-100bp discounts, as the STACR pool is likely to be, they could have seen higher defaults; however, we believe the tails would have driven a bulk of that and the clean tails in the STACR pool still bode well for defaults

Overall, based on Figure 7, it is easy to make the case that in the base case with average home prices running at 4-6% over the next 10 years, gross cumulative defaults should be capped at most at 1-1.2%. In fact, if anything, the tail characteristics shown above suggest that cumulative defaults could be lower than 1%. To get a sense of how the better collateral from the STACR deal should perform, we also look at historical jumbo FRM performance from the non-agency space below.

FIGURE 8

Historical default performance, HPA and tail characteristics of non-agency jumbo

Shorter of 10y from Orig. or until date				Orig. FICO Score		Original CLTV	
Vintage	Cum Def	Cum. HPA	Ann. HPA	Average	% FICO <680	Average	% OCLTV =>90
1999	0.7%	65%	5.1%	725	17%	75	0%
2000	0.9%	42%	3.6%	729	15%	76	0%
2001	0.4%	26%	2.3%	729	15%	75	0%
2002	0.4%	19%	1.7%	733	13%	73	0%
2003	1.5%	17%	1.6%	735	11%	73	1%
2004	3.2%	9%	0.9%	735	12%	75	2%
2005	9.8%	-11%	-1.5%	740	10%	76	4%
2006	15.5%	-19%	-3.0%	742	8%	78	8%
2007	18.6%	-17%	-3.1%	745	7%	80	12%
2008	15.8%	-13%	-2.6%	747	7%	78	5%
STACR 13-DN1				766	2.6%	76	3.4%

Note: Shows performance of the jumbo FRM non-agency universe limited to 60 to 80 OLTV loans. Defaults are defined as per the STACR Credit Events definition. Source: Corelogic, 1010data, Barclays Research

Non-agency jumbo performance hints at lower defaults on STACR

Figure 8 shows non-agency jumbo FRM performance by limiting the cohorts to 60-80 LTV. In FICO terms, the STACR collateral still looks better by about 30-40 points in average FICO versus the non-agency data from early 2000s vintage and by 10-15% in terms of the tails. While the CLTV tails in the early vintages seem very clean, this is likely from the fact that CLTVs and 2nd liens were not reported until recently on the non-agency side. As a result, we find that the jumbo collateral is likely somewhat cleaner versus the agency collateral from the early 2000s vintages but not by as much as the STACR reference pool. For these cohorts, we see that cumulative defaults over 10 years have been 0.5-1% for 2-4% annual HPA experience.

Overall, both agency and non-agency data point to defaults in the 0.5-1.2% range for collateral that was worse from a tail characteristics perspective but probably had higher prepaes working in favor and reducing the loans at risk of defaults when home prices fell

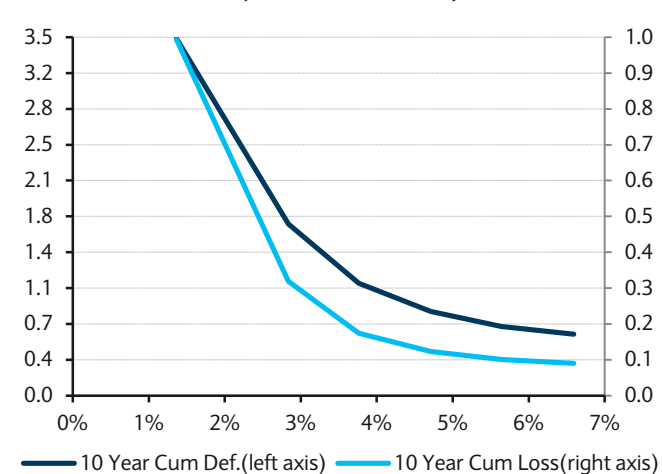
FIGURE 9

Cumulative defaults across economic scenarios

Scenario	Cum. HPA till 2023	Ann. HPA till 2023	Lifetime Cum Default	10 Year Cum Default	10 Year Cum Loss as per STACR
Down 20	16%	1.4%	3.9	3.5	1.0
Severe Stress	36%	2.8%	1.9	1.7	0.33
Stress	50%	3.8%	1.3	1.1	0.18
Base	66%	4.7%	1.0	0.8	0.12
Recovery	83%	5.6%	0.8	0.7	0.11
Strong Recovery	102%	6.6%	0.8	0.6	0.09

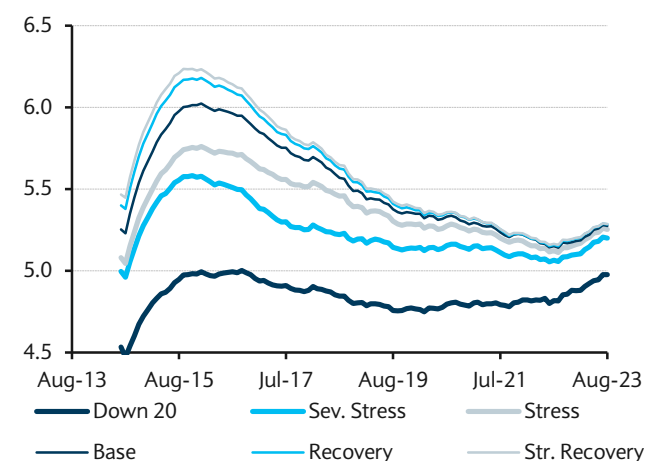
Source: Barclays Research

FIGURE 10

Cumulative defaults by annualized home prices until 2023

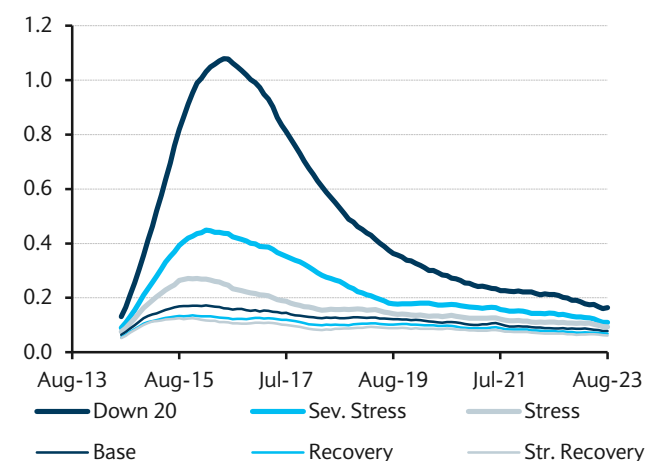
Source: Barclays Research

FIGURE 11

12 month average projected CRR across scenarios

Source: Barclays Research

FIGURE 12

12 month average projected CDR across scenarios

Source: Barclays Research

between 2006 and 2010. We believe that this forms a good range for base case defaults over 10 years for the STACR reference pool.

Model projections for cumulative credit events

Given the simple definition of liquidations and severities, investors need to model a more limited set of rolls than for non-agencies. The most important roll rates are likely to be current to prepaes, current to dq and dq to 180+ rolls. The model would also need to account for mods that happen before 180+ on these loans and cures. Severities do not need to be modelled because of the tiered severity scheme the deal uses.

If we tweak the Barclays Loan Transition Model to be better calibrated to the GSE loans level data, we get base case default projections that are roughly in the range we arrived at above by looking at 1999-2002 vintage Freddie and jumbo FRM data. Just to be clear, the model runs shown here were not estimated on the Freddie/Fannie loan level data but rather are based on calibrating the existing non-agency model to match more closely the GSE loan data. Figure 9 shows the projected cumulative defaults across scenarios over life and 10 years as well as the cumulative loss passed to the structure as per the STACR deals tiered severities. We also show the cumulative and annualized HPA over the period. Figure 10 shows the projected cumulative defaults and losses across the annualized HPA numbers. We find that, while defaults are fairly linear going from the recovery scenarios to base, when we run worse home price and unemployment scenarios the defaults increase much more dramatically. Because of the higher severities as defaults rise above 2% the profile for losses is even more convex. This should lead to fairly leveraged profiles on the underlying bonds, especially as we move to severe stress and down 20 scenarios.

Figure 11 and Figure 12 show the 1m average CRRs and CDRs for the STACR deal over the next 10 years. Given that most of the loans in the reference pool are 50-100bp discounts, we expect voluntary prepaes to remain fairly slow in the 5-7% range. The CDRs in the severe stress and down 20% scenarios go to 0.5-1% at the peak but the rest remain below 0.2% for the most part.

Offset from rep and warranty repurchases less meaningful

So far we have talked about our expectations for gross cumulative defaults given the collateral across a range of scenarios. However, as described in Section II, the deal also has features that protect borrowers from any defaults later found to be on loans with

underwriting defects or part of a rep and warranty settlement. So it is necessary to consider what offsets may be derived from this feature. Looking at historical data on this can be somewhat misleading, however. The underwriting standards fell dramatically in the 2004-2008 period and as a result, a much larger proportion of defaulted loans could be putback to originators than is likely on the reference pool for the STACR deal. The collateral in the reference pool is originated to much tighter underwriting standards; as a result, we believe it is less likely to be eligible for putbacks. Thus, we believe there should be only a fairly small offset from the rep and warranty feature, if any. Overall, we recommend against using a very aggressive number for repurchases when valuing these securities and, to be conservative, base our valuations on assuming that none of the loans gets repurchased.

Section III: Valuation

Given our base case expectations of collateral defaults and losses, these tranches look predictably attractive in the base case. However, given how thin and leveraged to losses these tranches are, valuation will have to consider some scenarios with much higher cumulative defaults to see how these tranches hold up. As a result, before diving into model scenario based valuations, we show cumulative defaults and M1/M2 bond losses across a range of CRR and CDR scenarios in Figure 13. Looking across the scenarios we find that the M1 survives with no bond loss until defaults reach 6-7%, depending on how high prepaids are run. These are almost 6-8 multiples vs. our base case default projection. On the other hand, M2 starts to take losses when defaults start to exceed about 1.5-2%.

FIGURE 13

Cumulative defaults and bond losses across CRR/CDR scenarios

	Cum Defaults				M1 Bond Loss				M2 Bond Loss			
	CDR				CDR				CDR			
CRR	0.1	0.25	0.5	1	0.1	0.25	0.5	1	0.1	0.25	0.5	1
2.5	1.3	3.3	6.4	12.3	0%	0%	0%	74%	0%	7%	64%	100%
5	1.1	2.6	5.2	9.9	0%	0%	0%	50%	0%	3%	51%	100%
7.5	0.9	2.1	4.2	8.2	0%	0%	0%	29%	0%	0%	40%	100%
10	0.7	1.8	3.5	6.9	0%	0%	0%	11%	0%	0%	31%	100%

Source: Barclays Research

The primary risk on the M1 is extension

Given that the M1 survives about 6-7% cumulative defaults before taking a loss and since this is a really clean reference pool, we believe that to get defaults in excess of 6-7%, another 2006-10 type event is likely needed. As a result, we believe that the primary risk on the M1 bonds is extension. This extension is a result of the triggers in the deal. As the losses pile up and enhancement falls, the triggers in the deal will start to fail. This, in turn, will lock out the M1 from receiving prepaids, and as a result could lead to substantial extension. Figure 14 shows stylized scenarios (running increasing CDRs at 5% voluntary prepaids) to depict the M1 bond losses and average life across a large range of defaults. We see that the M1 takes large losses only when defaults approach 10% on the collateral. However, as defaults trend to about 5%, the M1 starts extending out from being a 3-4y bond to a 7-8y bond due to failing triggers (M1 still receives the entire scheduled principal on the pool that is allocated to the M-1/M-2/B-H tranches).

FIGURE 14

M1 extends out due to triggers

CDR	0.25	0.5	0.75	1	1.25	1.5	1.75	2
Collat Default	2.6	5.2	7.6	9.9	12.2	14.4	16.5	18.5
Write-down	0%	0%	1%	50%	78%	83%	86%	87%
Average Life	4.6	7.6	7.9	7.4	6.0	4.9	4.2	3.6

Source: Barclays Research

M1 extension likely coincides with overall spread widening

While extension on the M1 is the primary risk, the bond is a floater so it can be argued that this extension risk should be limited. However, the scenario in which the M1 starts to extend is also likely to be a scenario in which spreads widen. If this coincides with an overall weaker economy, it may also be a scenario where rates stay low and the compensation that we see along the forwards is not realized. Due to the combined effect of extensions and spread widening, the M1 bond can have significant MTM losses in such a scenario. From a cash flow perspective, however, the M1 should be much better protected and as a result looks attractive on an unleveraged basis.

Triggers protect Freddie Mac from tail risks

The triggers in the deal that cause the M1 to extend are designed to protect Freddie Mac from a scenario of higher prepaes and higher defaults similar to what unfolded in 2006-10. Such a scenario would erode its g-fee stream and hit it with defaults at the same time. In the absence of triggers, the M1 would have paid down much faster and, hence, provide reduced credit protection for Freddie Mac. With triggers, Freddie Mac gives up additional g-fees since it pays the M1 coupon for longer on a larger balance, but it gets compensated in scenarios with very high defaults since more of the M1 is outstanding to absorb losses.

FIGURE 15

Model unleveraged valuations

Bond	Measure	Economic Scenarios						
		Down 20 (2x Def)	Down 20	Sev Stress	Stress	Base	Recovery	Str. Recov.
M1	Bond Loss	75	0	0	0	0	0	0
	WAL	4.8	7.0	4.5	4	3.7	3.6	3.6
	Libor ZV	-2247	340	340	340	340	340	340
	Cum CF %	50	142	124	120	118	118	117
M2	Bond Loss	100	51	1.3	0	0	0	0
	WAL	2.9	7.5	10	9.9	9.8	9.8	9.8
	Libor ZV	-7143	61	707	715	715	715	715
	Cum CF %	23	122	199	199	199	198	198
Collateral	Defaults	7.9	3.48	1.68	1.1	0.82	0.67	0.6
	Losses	2.76	0.99	0.32	0.17	0.12	0.1	0.09

Source: Barclays Research

M2 has positive yields until about 3.5-4% cumulative defaults

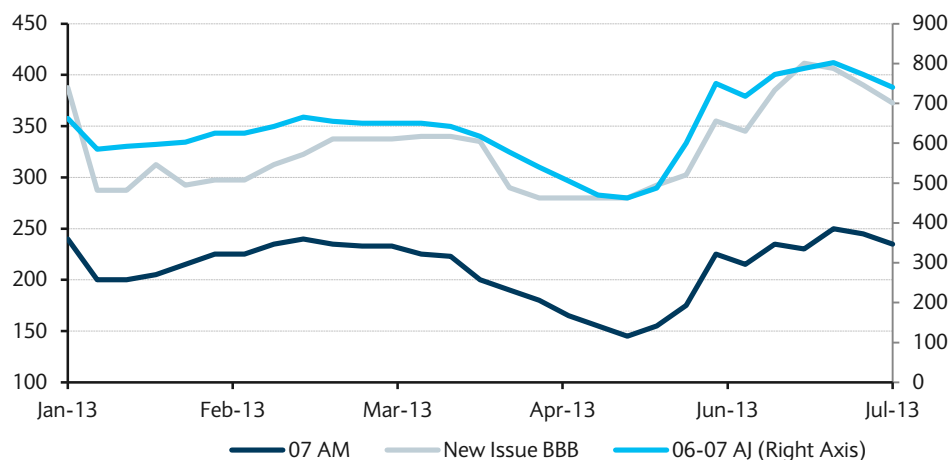
Next we look at model valuations across the six scenarios as well as an extra stressful scenario in which we knob the model to run at twice the current-to-delinquent roll rates in the down 20% scenario (Figure 15). We find that the M1 survives in all but the most stressful scenarios, and as we pointed out above, the main risk is extension. On the M2, however, which has only 0.3% credit support, losses start to hit the tranche as cumulative defaults start to approach about 1.5-2%. Despite this, because of the high coupon offered on the bond (L+715bp), the

yield stays positive until defaults reach about 3.5-4% in the down 20% scenario. While the 3-4% defaults number seems like a fair cushion given that even the much worse 2003-2004 vintage Freddie Mac cohorts saw lower losses with a much worse housing scenario, these older vintage cohorts had the benefit of a few years of fast prepayments before the economy turned. As a result, a smaller factor was outstanding on the pool that had to go through the worse economic periods and that likely led to lower defaults. Since the reference pool loans are mostly 50-100bp discounts, prepaes on this pool are unlikely to pick up much even if home prices increase significantly; as a result, along the forwards prepaes should remain at 5-7 CPR. Thus in a poor economic scenario, we will have a much higher factor outstanding and at risk than the 2003-2004 cohorts so getting to a 3-4% default number is not as unlikely as it may at first seem. As such, we believe that on an unleveraged basis the M2 looks attractive only for investors convinced that the chances of another major downturn are very small.

Comparables trading somewhat tighter

Looking at the profiles in a vacuum, it is hard to see whether these bonds are rich or cheap, so we need to look for similarly leveraged profiles as comparison points. There is very little price discovery in the new issue subordinates in the RMBS space since the largest issuer (i.e., SEMT) is a REIT that retains most of the subordinate pieces. Looking across to the CMBS space, we can compare new issue BBB tranches or seasoned AM/AJs to the M1/M2s. Naturally the comparison cannot be exact but it can be argued that the M1s are closer to new issue BBBs or 07 AMs, while M2s are closer to AJs in 2006-07 vintage (Figure 16). The CMBS bonds have more idiosyncratic risk, due to the chunky nature of the underlying collateral in CMBS where a single loan could potentially make or break these bonds. So it can be argued that the STACR bonds should trade tighter because even with the thin tranches these have very little idiosyncratic risk given the approximately 100,000 loans in the reference pool with fairly good diversification across geographies.

FIGURE 16
Comparable spreads from the CMBS universe



Source: Barclays Research

Leverage makes the profiles seem even more attractive.

These bonds can be purchased with leverage, unlike the CMBS bonds we are using for comparison, so it becomes important to look at the economics from a leveraged standpoint as well. Figure 17 shows leveraged yields under these scenarios assuming a 20% haircut on M1 and 40% on M2 and assuming L+150bp funding cost, with funding assumed to be available until the bond is outstanding. We also assume that investors post margin as required to maintain the leverage ratios under all scenarios, since the leverage available in most cases is in recourse form. The leveraged yields in the base and stress case on the M1 are extremely attractive and can be thought of as adequate compensation against the worst

scenarios (Figure 17). So while buying these tranches in leveraged form entails additional risk in the form of MTM volatility risks, we believe that the M1 looks especially attractive with double-digit yields, and losses remain very unlikely, except in case of extreme MTM volatility. However, given our overall outlook for housing in the medium term, we would be very comfortable owning that risk.

The M2 is inherently more leveraged and using repo financing amplifies that even further. In fact, given how low prepaes are likely to be along the forwards on this collateral, the M2 probably gets no principal until the 10y maturity, so these are in most cases 10-year bullets from the point of view of principal. However, given that it is such a thin tranche, the downside (which is most likely the risk of being wiped out completely) does not get amplified as much relative to the upside. For instance, look at the cumulative cash flows as a fraction of the upfront cost in unleveraged and leveraged forms on the M2 (Figure 16 and Figure 17). In unleveraged form, until losses are at the down 20% case, investors receive 200% of their initial investment in cash flows over life, while in the down 20% case, it falls to 120%. In leveraged form, in the down 20% case, investors will still cover their initial investment (with a small positive yield). In scenarios better than the severe stress, however, they will get back about 280% of their initial equity investment. As such, an 17-18% base case leveraged yield is attractive relative to any similarly leveraged instrument. As a result, in leveraged form, it appears that the M2 bonds are compensated for the credit risks and the leveraged profile.

FIGURE 17
Model leveraged valuations assuming 20/40% haircuts for M1/M2, respectively

Bond	Measure	Economic Scenarios						
		Down 20 (2x Def)	Down 20	Sev Stress	Stress	Base	Recovery	Str. Recov.
M1	C/F AL	-	6.0	4.1	3.6	3.4	3.3	3.3
	Yield (%)	-ve	12.9	12.4	12.3	12.2	12.2	12.1
	Cum CF %	6	196	159	150	147	145	145
M2	C/F AL	-	6.5	6.9	6.9	6.8	6.8	6.8
	Yield (%)	-ve	1.5	17.5	17.6	17.6	17.6	17.6
	Cum CF %	-ve	110	283	285	283	282	282
Collateral	Defaults	7.9	3.48	1.68	1.1	0.82	0.67	0.6
	Losses	2.76	0.99	0.32	0.17	0.12	0.1	0.09

Note: We assume funding spread of 150bp and haircuts of 20/40% on M1/M2 respectively. Source: Barclays Research

Are the risks hedgeable?

Investors in these bonds are making a call on the underlying mortgage credit fundamentals and, as a result, may not want to hedge that risk (not that there are too many obvious hedges for this tail risk). However, given the leverage in these tranches to spread risk, especially in an extension scenario, it may make sense to hedge some of the spread risk using instruments such as HY CDX, particularly when using leverage.

The rate duration of these instruments is harder to pin down since it remains unclear whether rising rates coincide with an improving economy and tighter spreads or whether rising rates are a result of the Fed stopping QE, and hence they coincide with wider spreads. If we make the first assumption, as the loans are already discounted by 50-100bp, prepaes will probably remain range bound at 5-7% CRR. On the other hand, if the economy is improving, that could improve default estimates and in a tighter spread environment may actually lead these bonds to trade with negative rate duration. On the other hand, if rates

sell off with wider spreads, then these bonds will trade to longer positive durations. As such, any rate hedges would have to be a function of taking a view on the correlation between rates and risk premiums, and those would have to necessarily be based on short-medium term rate views. As a result, we find it hard to recommend hedging these bonds for rates given that the hedge would have to be viewed as a separate trade.

Potential for CDS/CDX market on these tranches

While Dodd/Frank has made it harder to trade synthetic instruments, if this becomes a more regular program of issuance, then it is likely ABX/CMBX style CDX indices will crop up on these transactions. If such a market does get established, then it would be a great avenue for issuers in private label deals and warehouse lenders to hedge the mortgage credit spread risk.

Section IV: Future prospects and sizing the market

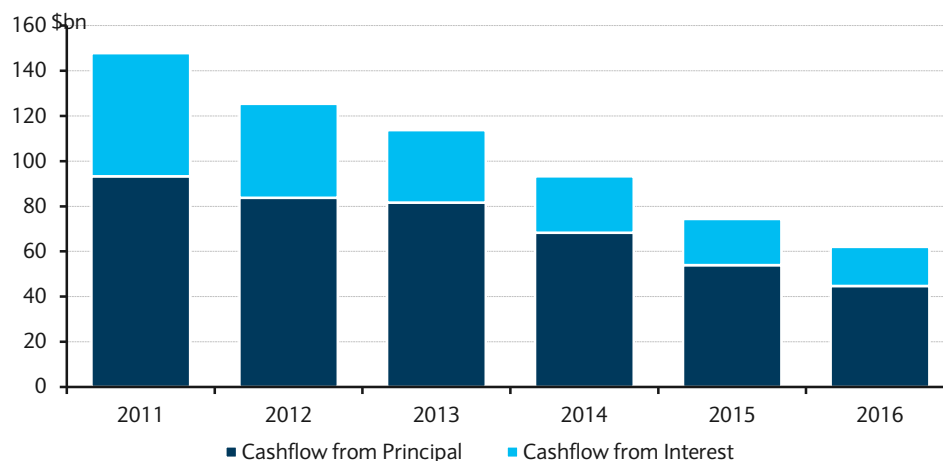
The natural question to ask when considering whether this can become an ongoing issuance program is about market sizing. How much credit supply are we talking about? For now Freddie Mac has issued a deal on the 60 to 80 LTV bucket of its Q3 2012 issuance. If both GSEs get involved and start selling all the credit risk of their issuance through this program, we could see \$20-25bn of issuance annually (assuming \$600-800bn of gross mortgage supply and a tranche size of 3% in aggregate). While it is unlikely that this would happen this year or even in 2014, we believe that eventually there could be enough capacity in the sector to absorb this supply.

Where will the demand come from?

One big source of potential demand is existing non-agency MBS investors. There is currently about \$900bn (face value) outstanding in the non-agency market. This principal is paying down at about \$80bn per year (a combination of prepayments and liquidation recovery cash flows). Figure 18 shows our estimates of the future cash flow from the non-agency MBS universe by principal and interest. We expect principal cash flows of nearly \$80bn over 2013, another \$70bn in the year after and even higher amounts if interest is included. Admittedly, some of that is going to legacy mortgage holders, who will not re-invest into credit leveraged bonds. But even if one assumes that 20-25% of what is paid down is re-invested, we are looking at \$15-20bn of demand simply from existing mortgage credit investors.

On top of that, we expect some new demand as well. While the structure may present some initial difficulties, we believe that for this program to be sustainable, REITs, insurance companies and money managers will have to play a much bigger role in the space. As shown in the section on valuation, the yields in both leveraged and unleveraged form should make the investment attractive to these investors. Money managers have historically been more interested in higher-rated securities but have become more comfortable investing in lower-rated non-agency securities over the past few years and that expertise can be put to use with this market. We could also see new capital being raised through REIT-like vehicles for this product, if the tax treatment works for them. As of December 2006, Fannie Mae and Freddie Mac had market capitalizations of about \$100-130bn dollars combined, and while this new model may not have the same leverage as the Fannie Freddie model, there should be continued investor appetite for equity-like profiles based on mortgage credit. Thus, we believe that there will be enough demand to absorb the issuance of these credit tranches over the medium term.

FIGURE 18

Principal pay-down on existing non-agency investors

Source: Barclays Research

Protecting the taxpayer

The GSEs have historically hedged much of their interest rate risk in the interest rate markets. But credit risk has always been much bigger, and they have historically not hedged that at all, with negative consequences for the taxpayer. One can argue that the government should be in the business only of providing catastrophe insurance, stepping in when the private sector cannot, and should not be carrying the first-loss risk. Transactions like the STACR deal start the process of transferring that risk away from taxpayer-supported entities such as Freddie Mac.

The M1/M2 priced to a combined spread of 527.5bp for the 0.3-3% tranche. This translates to about 14.25bp running in g-fee terms on the loan. The average g-fee for these loans (net of the 10bp going to Congress) is probably on the order of 30-35bp. As a result, the excess g-fee that Freddie retains in this deal should be more than enough to cover the first 30bp of loss as well as any set-up a reserve for any tail risks that may remain.

Risk-based pricing of the guarantee business

One of the most important benefits of this approach is the market-based signal it sends to the GSEs. In the ideal scenario, the GSEs should be able to sell the equity tranche at par by simply passing on part of the guarantee fee (g-fee) they are charging on the loans to the equity tranche. These deals allow for a simple feedback mechanism to the GSEs on the g-fee they are charging and allows them to reprice the g-fee when the deal economics so dictate.

Alternately, the GSEs/administration could decide to continue making these loans at the same g-fee for social/policy reasons to counteract cyclical forces. But through this mechanism, policymakers will at least have a way to estimate the actual dollar amount of the subsidy being given by the taxpayer to borrowers and will allow them to make a more informed decision on whether the benefits of such a subsidy outweigh the costs.

Helping housing finance restart

Such a program can help housing finance on both the agency and non-agency mortgage fronts. A primary market needs a healthy, liquid secondary market. And secondary markets need clearly defined benchmarks against which to trade, which is why indices exist. Similarly, once the first-loss pieces of a series of well-defined, liquid, and large cohorts start trading, it should provide a series of benchmarks against which the non-agency market can price the credit part of new issuance.

On the agency mortgage front, a program like this might provide Congress with a way forward on Freddie Mac and Fannie Mae. After all, if the credit risk in ongoing agency mortgage origination is being smoothly transferred to the private sector, the urgency to wind down the GSEs could decrease considerably and more efficient structures can be explored.

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